

MSMS 401 : Bayesian Statistics and Computation

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⊕ Problem

Let $X_i \stackrel{\text{iid}}{\sim} \text{Geometric}(\theta)$ on support $\{0, 1, 2, \dots\} \forall i = 1(1)n$ and *a priori* $\theta \sim \mathcal{B}_1(\alpha, \beta)$, both α and β are unknown. Obtain empirical Bayes estimate of α and β by MLII method.

⊕ Solution

$$f(x | \theta) = \theta(1 - \theta)^x, \quad x = 0, 1, 2, \dots, \quad 0 < \theta < 1.$$

$$\pi(\theta; \alpha, \beta) = \frac{1}{\mathcal{B}(\alpha, \beta)} \cdot \theta^{\alpha-1}(1 - \theta)^{\beta-1}, \quad 0 < \theta < 1.$$

Then the joint marginal likelihood will be

$$\begin{aligned} L(\alpha, \beta) &= \int_0^1 \prod_{i=1}^n f(x_i | \theta) \cdot \pi(\theta; \alpha, \beta) d\theta \\ &= \int_0^1 \prod_{i=1}^n \theta(1 - \theta)^{x_i} \cdot \frac{1}{\mathcal{B}(\alpha, \beta)} \cdot \theta^{\alpha-1}(1 - \theta)^{\beta-1} d\theta \\ &= \int_0^1 \theta^n (1 - \theta)^{\sum_{i=1}^n x_i} \cdot \frac{1}{\mathcal{B}(\alpha, \beta)} \cdot \theta^{\alpha-1}(1 - \theta)^{\beta-1} d\theta \\ &= \frac{1}{\mathcal{B}(\alpha, \beta)} \cdot \int_0^1 \theta^{\alpha+n-1} (1 - \theta)^{\beta + \sum_{i=1}^n x_i - 1} d\theta \\ &= \frac{1}{\mathcal{B}(\alpha, \beta)} \cdot \mathcal{B}\left(\alpha + n, \beta + \sum_{i=1}^n x_i\right) \\ &= \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} \cdot \frac{\Gamma(\alpha + n)\Gamma\left(\beta + \sum_{i=1}^n x_i\right)}{\Gamma\left(\alpha + \beta + \sum_{i=1}^n x_i + n\right)} \end{aligned}$$

$$= \frac{\Gamma(\alpha + \beta) \cdot \Gamma(\alpha + n) \cdot \Gamma\left(\beta + \sum_{i=1}^n x_i\right)}{\Gamma(\alpha) \cdot \Gamma(\beta) \cdot \Gamma\left(\alpha + \beta + \sum_{i=1}^n x_i + n\right)}$$

The log-likelihood will be

$$\begin{aligned} \ell(\alpha, \beta) = & \ln \Gamma(\alpha + \beta) + \ln \Gamma(\alpha + n) + \ln \Gamma\left(\beta + \sum_{i=1}^n x_i\right) \\ & - \ln \Gamma(\alpha) - \ln \Gamma(\beta) - \ln \Gamma\left(\alpha + \beta + \sum_{i=1}^n x_i + n\right). \end{aligned}$$

Here both α and β are unknown. Partial differentiation of $\ell(\alpha, \beta)$ w.r.t. α and β introduces **digamma** function. Hence, analytical solution of the equations

$$\frac{\partial}{\partial \alpha} \ell(\alpha, \beta) = 0$$

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is not possible and we resort to numerical methods for approximate solutions.